Detection of Event of Interest for Satellite Video Understanding

Yanfeng Gu, *Senior Member, IEEE*, Tengfei Wang, *Student Member, IEEE*, Xudong Jin, *Student Member, IEEE*, and Guoming Gao, *Member, IEEE*

***Abstract*— Satellite videos provide rich dynamic information of observed scenes at a large spatial and temporal scale and will play an important role in the future space information network. This work devotes to revealing events of interest (EOI) from satellite video scenes by using a two-stream method. In satellite videos, individual frames reflect the static information like the basic scenes where the event was happening, while a sequence of frames determines the motion information. Considering these facts, a novel two-stream EOI detection framework is proposed, where one stream extracts static spatial information of satellite videos by AlexNet, whereas the other stream extracts the motion information using a local trajectories analysis method. First, the whole video scene is segmented into small spatial-temporal patches, where labeling EOI and non-EOI is completed. Next, the trajectories are extracted from 3-D satellite video cubes that are generated from event scene patches. Finally, this trajectory classification process is treated as a weak supervision learning problem and solved by sparse dictionary learning. The experimental results demonstrate that the proposed two-stream method is effective for EOI detection and has a huge potential for satellite video scenes analysis and understanding. The proposed method also outperforms the existing competitive models for video analysis.**

***Index Terms*— Event detection, event of interest (EOI), satellite video, two-stream framework, video understanding.**

I. Introduction

SATELLITE videos open a new viewpoint to observe the Earth in the field of remote sensing. They provide rich motion information that can help us better understand the events that happened on the ground (e.g., traffic jam, the status of airport, and harbor).

Most satellite videos acquired by low orbit satellite have a high spatial resolution which ranges from 0.8 to 2 m, as well as a frame rate ranging from 4 to 30 (although some videos have a frame rate less than 25, the objects' moving trajectories are still continuous, and they are considered as videos). The important targets' motion of cars, ships, and planes can be clearly observed. Combined with multisatellite system, it may put out a surge of energy on geoscience and remote sensing science.

Satellite video processing has drawn much attention to the potential applications in the future space information network. Many research studies have been launched on satellite videos. The existing research studies mainly focus on two aspects: One aspect is applying existing processing methods such as object detection and tracking that have been the subject of many typical studies in video processing [1]–[5]. The other aims to enhance the satellite videos by raising the spatial resolution so that subsequent processing methods can achieve better performance [6]. They have been found to be useful for analyzing satellite videos but are limited to process local targets. Considering the large scale of satellite videos, it is challenging either to track thousands of targets in a video or to raise spatial resolution. In practice, some targets are too small to be detected easily, which greatly increases the complexity of satellite video processing and sharply narrows down its application. In view of this, it is more helpful to find a new way to process videos of large scale: detect proper regions for video understanding.

In recent years, there has been an increasing interest in scene image processing which solves the problem of how to process large-scale images for image understanding. Revealing regions of interest (ROIs) from images is a common method to address the large-scale problem of remote sensing image processing. As regards satellite videos, the ROIs extracted from every individual frame would together form an "event scene," what we call the event of interest (EOI). The notion of EOI is similar to event detection [7], which aims at ranking a sequence of unconstrained videos according to the likelihood of belonging to a certain event. However, the goals of EOI and event detection are different: Event detection for common videos mainly focuses on temporal shots when the events happened, EOI for large-scale satellite videos focus on spatial regions where the events happened in a large area. The area is captured by video satellite in a period of time. In Fig. 1, an example is provided to illustrate the EOI. Generally, the regions are detected based on the similarity between spatial information and motion information. To the best of our knowledge, there is no previous study on EOI detection in satellite videos. The first attempt is made in this work.

Event detection, which can also be applied in EOI, is helpful and feasible to analyze the targets' motion for video

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The authors are with the Department of Information Engineering, Harbin Institute of Technology, Harbin 150001, China (e-mail: guyf@hit.edu.cn).

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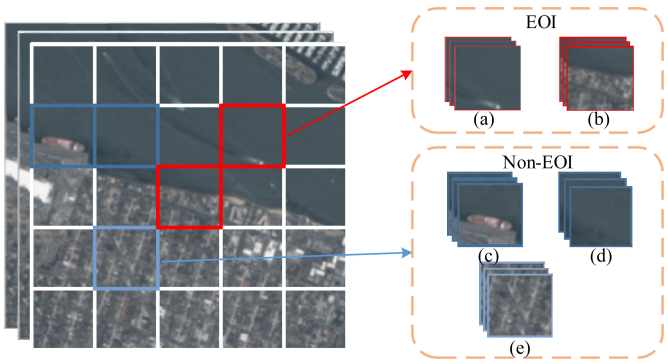


Fig. 1. Example to illustrate our task. Given a satellite video, a patch was regarded as an event. The EOI such as (a) ship sailing and (b) car running. The non-EOI such as (c) and (e) static events and (d) flowing water.

understanding. The current methods of event detection focus on video classification, using classification methods [8]–[12] to move apart the scenes of interest from the others. Some methods use event abnormal detection [13]–[15] for the abnormal events are necessary, including EOIs. However, most event detection methods find difficulty dealing with small targets' motion and videos' low signal-to-noise ratio (SNR) when applied in satellite videos. Most of the existing detection methods are based on the similar features between the same samples without considering the tiny motions, for example, small cars' motion would be ignored. The noise and video jittering in satellite videos influence the features extraction much more severely since the noise may drown the targets in one frame or some continuous frames, the situation usually that does not happen in traditional video processing. It is found that almost all the traditional methods have difficulty distinguishing the noise and targets, even optical flow [16], [17] cannot get a proper result. Hence, directly applying the methods of computer vision on satellite videos is impractical.

To solve the challenges in this task, a two-stream algorithm is proposed to reveal the underlying EOIs from the complex spatial–temporal patches. In this article, a large-scale video is segmented into small videos by spatial positions and generates a sequence of video scenes. At present, a typical satellite videos' duration is short, so a small video can be treated as an event. Hence, the problem of detecting EOI from large-scale video can be transferred into the problem of finding proper events from the small videos. Therefore, what this work wants to do is to detect the EOI from the small videos of event, where the moving scenes like those of sailing ship and moving cars are of special interest. Observing that the satellite videos have a huge range of static background and relatively small but massive objects, the redundant background information is separated from the involute but important motion information. As for detecting the EOI scenes from a huge video scene, the two-stream method gives an inspiration to deal with the static information and motion information separately [18]. The first single frames of each small videos are chosen to extract the background information. The frame differencing sequence is used to extract motion information by the spatial–temporal analysis. Fusing the static information and motion information creates a powerful representation for videos and benefits event understanding.

In the static scenes feature extraction stream, an AlexNet is used to extract the feature of background. It is a typical scene classification question. Deep learning [19], [20] methods have a great potential to deal with this question. Almost all the state-of-the-art methods recognize convolutional neural networks (CNN) as the best choice to deal with the scene classification question [21]–[26]. CNN obtains the best performance of classification due to millions of parameters and exhaustive samples. The complex scene may cause small sample question, and transfer learning [27]–[29] is seen as a method to solve this question. The AlexNet's origin labels are transferred as features of static scenes, for the detection scenes' labels, and background scenes' labels are overlapped, given the scenes where cars were moving on the road, for example, about 30% of them would be identified as EOI, while the remaining would be identified as non-EOI, although they would have the same label given by AlexNet.

In the other stream, a local trajectory analysis method is proposed to extract the motion features of events. A spatial–temporal analysis method [30], [31] is used to extract the trajectories of small targets' motion. Spatial–temporal method can detect the small moving targets' trajectories efficiently despite the influence of noise and jittering environment. To detect the trajectories of EOI's object motion, the trajectories of moving objects were segmented into local patches. The goal is using the features of local patches of an event to represent a whole event. Only events are labeled in the training step. The question to represent the events transfers to a weak supervision learning question. A sparse representation with dictionary learning method was used to solve this question.

Finally, the features of small videos that are extracted by spatial stream and temporal stream are inputted to a two-class support vector machine (SVM). To evaluate the proposed approach, the experiment was conducted on a collection of satellite videos data set. The proposed method is compared with a number of methods. Experimental results demonstrate that our method outperforms other compared methods. Comprehensive experiments on our data set have conducted to demonstrate the effectiveness and efficiency of the proposed method. The method would potentially benefit a variety of applications. For example, it can help researchers to quickly find the moving targets from the satellite videos instead of processing a whole video.

To sum up, this work has twofold contributions.

1) It is the first attempt to apply EOI detection in satellite videos to improve satellite video understanding. The challenges and their solutions are described.
2) A two-stream method that can solve large-scale question and low SNR question is proposed to detect EOI scenes.

The remainder of this article is organized as follows. The algorithm is presented in Section II. Data description is provided in Section III, the details of data set we had created are shown in this section. The experimental results in Section IV, we compared the baseline method with our proposed method. Finally, Section V concludes the whole article.

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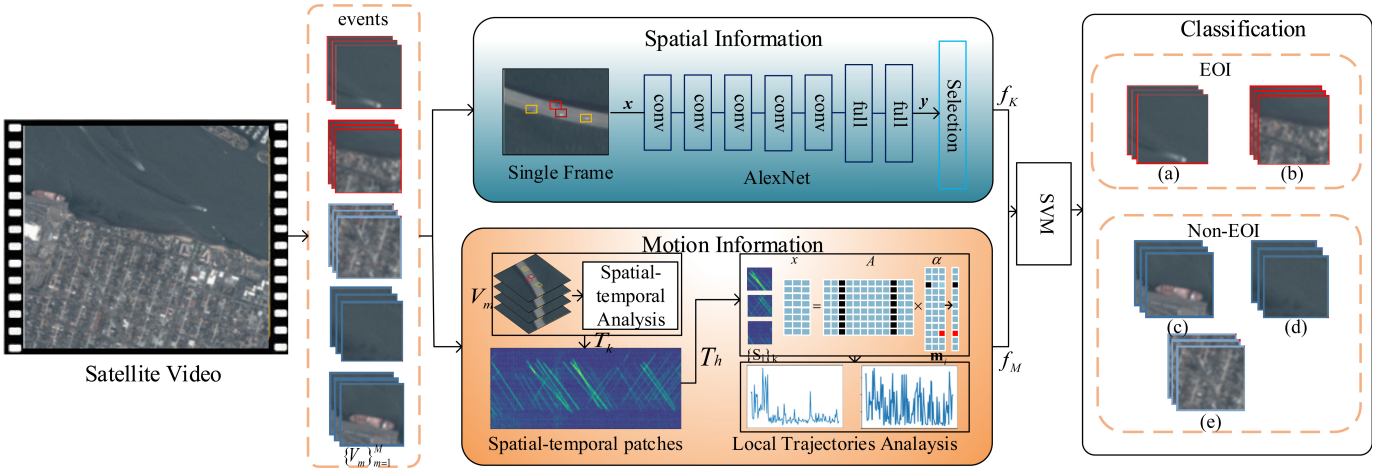


Fig. 2. Main framework of EOI detection for satellite videos understanding.

II. Proposed Approach

A. Algorithm Overview

Fig. 2 shows the framework of the proposed approach. Suppose a huge satellite video can be segmented into M small videos $\{V_m\}_{m=1}^{M}$ or called M events. The goal is to find EOI from those M events by predicting the scores of interests that reflect the probabilities of each event. A two-stream method that contains spatial stream and temporal stream is used to extract the static features f_K and motion features f_M. In the spatial stream, the spatial information is extracted from the first frame of each video $\{V_1\}_{m=1}^{M}$. An AlexNet is used to extract the static information f_K by transfer learning. The parameters of AlexNet that extracts hundreds of features from an image are transferred from the training result of Pascal VOC data set. For the condition of EOI detection, only features that are selected by calculating the probabilities of training samples are useful. Those static spatial features are represented as f_K.

Motion information is one of the most important aspect in the video processing field. In the motion stream, for an event V_m, the trajectories of targets' moving are extracted by the spatial—temporal analysis from the event and represent as a spatial–temporal slice. Then a trajectory T_k is segmented into local patches $\{S_i\}_k$ to solve the locating problem. Because the whole event' labels are known in training step, it means the label of whole local patches $\{S_i\}_k$ is known, but the label of a patch S_i is unknown. Representation of the trajectories by the local patches is a weak supervision problem. A sparse dictionary learning (SDL) method is used to solve this question and gets the motion feature f_M. An event is represented by combining the motion features f_M and static features f_K. At last, SVM is used to classify the EOI and non-EOI.

B. AlexNet and Feature Transferring

Generally, an AlexNet includes five convolutional layers, three max-pooling layers, and two soft max layers. In Fig. 3, the simple representing graph of AlexNet's structure is provided. It uses a backward method to adjust the thousands of parameters and had obtained a better result in complex scenes classification. An AlexNet includes two steps of training and testing. Suppose $\boldsymbol{x}$ is the input and $\boldsymbol{y}$ is the output, it uses the cost function [32], [33] as follows:

$$E^N = \frac{1}{2}\sum_{n=1}^{N}\sum_{k=1}^{c}\left(t_k^n - y_k^n\right)2 \tag{1}$$

where E^N is the cost of whole categories. y_k^n is the wanted output of the nth layer. t_k^n is the prediction of y_k^n, an AlexNet have N layers, c is the number of categories. y^N is the final output, and t^N is the wanted output.

In testing step and training step, y^{l-1} is obtained by the feedforward propagation of convolutional layers and pooling layers. Where the $(l-1)$th layer's output y^{l-1} is the lth layer's input x^l, and the first layer's input is the image. In convolution layers and pooling layers, the forward propagations are different.

The forward propagation of convolutional layers is

$$x^l = f'(u^l), \quad u^l = W^l x^l - 1 + b^l. \tag{2}$$

The forward propagation of pooling layers is

$$x_j^l = f\left(\beta_j^l \mathrm{down}\left(x_i^{l-1}\right) + b_j^l\right) \tag{3}$$

where l is the layer number. u is the output of a convolutional layer. W is the weight, b is the bias of the linear function. y^N is obtained by the forward propagation step by step.

n the training step, W and b of each layer are obtained by back-propagation from the cost E^N of the Nth layer.

The back-propagation pass is

$$\frac{\partial E}{\partial b} = \frac{\partial E}{\partial u}\frac{\partial u}{\partial b} = \delta \tag{4}$$

$$\delta^l = (W^{l+1})^{\mathrm{T}}\delta^{l+1} \circ f'(u^l). \tag{5}$$

We have

$$\frac{\partial E}{\partial W^l} = x^{l-1}(\delta^l)^{\mathrm{T}} \tag{6}$$

$$\Delta W^l = -\eta\frac{\partial E}{\partial W^l}. \tag{7}$$

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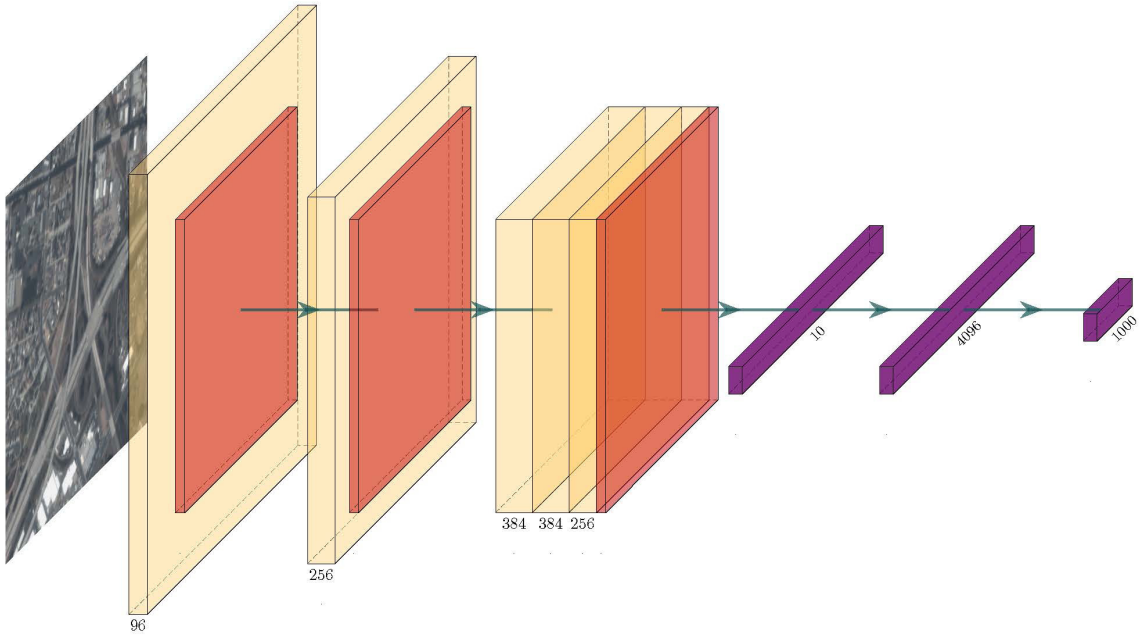


Fig. 3. Structure of AlexNet.

Also, the back-propagations are different in convolutional layers and pooling layers. In the convolutional layers, the formulation of parameters passing, where up(x) is the upsampling of x, β_j^{l+1}

$$x_j^l = f\left(\sum_{i\in M_j} x_i^{l-1} * k_{ij}^l + b_j^l\right) \tag{8}$$

$$\delta_j^l = \beta_j^{l+1}\left(f'\left(u_j^l\right)\right) \circ \mathrm{up}\left(\delta_j^{l+1}\right) \tag{9}$$

$$\mathrm{up}(x) \equiv x \otimes 1_{n\times n} \tag{10}$$

$$\frac{\partial E}{\partial b_j} = \sum_{u,v}\left(\delta_j^l\right)uv \tag{11}$$

$$\frac{\partial E}{\partial k_{ij}^l} = \sum_{u,v}\left(\delta_j^l\right)uv\left(p_i^{l-1}\right)uv. \tag{12}$$

In the pooling layers, the formulation of parameters passing

$$x_j^l = f\left(\beta_j^l \mathrm{down}\left(x_i^{l-1}\right) + b_j^l\right) \tag{13}$$

$$d_j^l = \mathrm{down}\left(x_j^{l-1}\right) \tag{14}$$

$$\frac{\partial E}{\partial \beta j} = \sum_{u,v}\left(\delta_j^l \circ d|_j^l\right)uv. \tag{15}$$

The proposed method transfers only the parameters of AlexNet that was training on the Pascal VOC data set. The labels of AlexNet are transferred as the static scenes' features since it has been proven to obtain a better result in remote sensing scenes classification and only need to adjust the fully connected layers. Considering our data set scenes, an AlexNet is used on the whole videos' first frames. Statistic shows that almost all the scenes (about 90%) focus on f_K classes. So, the most representative labels are transferred as the static spatial features.

C. Spatial–Temporal Analysis

Changes in image intensity are only due to translation of local image intensity in satellite videos, and the object motion [30] can be formulated as follows:

$$I(x+u_x(x,y,\phi),\ y+u_y(x,y,\phi),\ t+1) = I(x,y,t) \tag{16}$$

where $I(x,y,t)$ is the pixel intensity of location (x,y) in frame t.

Suppose a video object at position $(x,\ y)$ moves from frame t to $t+1$; then, the motion model [28] with parameter $\phi = [a_1, a_2, a_3, a_4, d_x, d_y]^{\mathrm{T}}$ is

$$u(x,y,\phi) = \begin{bmatrix} u_x(x,y,\phi) \\ u_y(x,y,\phi) \end{bmatrix} = \begin{bmatrix} a_1 & a_2 \\ a_3 & a_4 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} d_x \\ d_y \end{bmatrix} \tag{17}$$

where a_1, a_2, a_3, a_4 is the rotation parameters, and d_x and d_y are the translation parameters.

From the first-order Taylor series expansion, we know that

$$I_t(x,y) + [I_x(x,y), I_y(x,y)] \cdot u(x,y,\phi) = 0. \tag{18}$$

And

$$u(x,y,\phi) = \frac{-I_t(x,y)}{[I_x(x,y), I_y(x,y)]} = \frac{-I_t(x,y)}{I_s(x,y)}. \tag{19}$$

Equations (1)–(4) show that the objects moving determined only by $I_s(x,y)$ and $I_t(x,y)$. A spatial–temporal slice is a 2-D image consisting of rows of pixels [29] that transforms an *XYT* video space into an *XT* image space, thus it achieves a debasing method from video to image. It is similar for horizontal or for vertical, so we get the horizon direction as the example. For a video $V(x,y,t)$, suppose the video length is L, and the frame size is $H \times W$, then the horizontal spatial–temporal slice [27],

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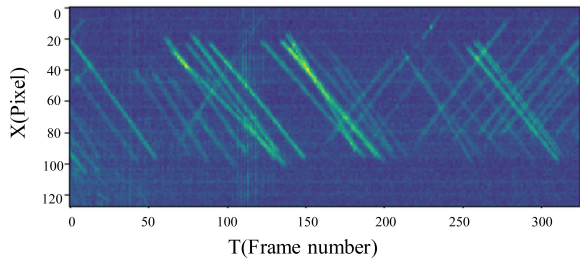


Fig. 4. Trajectories of some running cars in X–T.

S_k can be formulated as follows:

$$S_k = \begin{bmatrix} i_1^1 & \dots & i_i^1 & \dots & i_L^1 \\ \vdots & & \vdots & & \vdots \\ i_1^j & \dots & i_i^j & \dots & i_L^j \\ \vdots & & \vdots & & \vdots \\ i_1^W & \dots & i_i^W & \dots & i_L^W \end{bmatrix}_{y=k}, \quad k \in [1, H] \tag{20}$$

where T_k is the difference of S_k

$$T_k = \begin{bmatrix} i_2^1 - i_1^1 & \dots & i_{i+1}^1 - i_i^1 & \dots & i_L^1 - i_{L-1}^1 \\ \vdots & & \vdots & & \vdots \\ i_2^j - i_1^j & \dots & i_{i+1}^j - i_i^j & \dots & i_L^j - i_{L-1}^j \\ \vdots & & \vdots & & \vdots \\ i_2^W - i_1^W & \dots & i_{i+1}^W - i_i^W & \dots & i_L^W - i_{L-1}^W \end{bmatrix}. \tag{21}$$

The trajectories in $X-T$ slice T_h are obtained by calculating the average of T_k that

$$T_h = \overline{T_k}, \quad k = 1, 2, \dots, H. \tag{22}$$

In Fig. 4, the trajectories of some running cars in X–T slice are illustrated. Noise and video jittering may influence the trajectories, but the dimensionality is from a 3-D cube to a 2-D image. Generally, the trajectories of moving objects are focused on in a time, so it is easy to detect the moving targets' trajectories than processing a 3-D satellite video cube.

D. Weak Supervision Learning

Weak supervision learning [33]–[35] for object detection is similar to self-training although the training data are not completely labeled. The question of weak supervision learning is of interest because in fact most of the data are hard to label. Some semisupervision learning research studies [35]–[37] are helpful for this question. In our approach, the EOIs are wanted to be detected from the spatial–temporal slices, but the trajectories of moving targets are not in a fixed location of the slices. It is difficult to extract the features from a whole image that only contains background, noise, and unfixed targets. So, the slices were segmented into local patches to solve the locating problem. In the training step, the labels of events and spatial–temporal slices are the same. Assume the training sample of spatial–temporal slices are $[T_1, T_2, \dots, T_n]$, the patches of slice are $[\{S_i\}_1, \{S_i\}_2, \dots, \{S_i\}_n]$. We only know the labels of T_k, if T_k is the slice of EOI, then $\{S_i\}_k$ may contain the target patches, noise patches, and jittering patches. Finding EOI from the events becomes finding spatial–temporal slices by local patches' features, but in training step, only events' labels are given. So, the question of EOI detection becomes a weak supervision learning question.

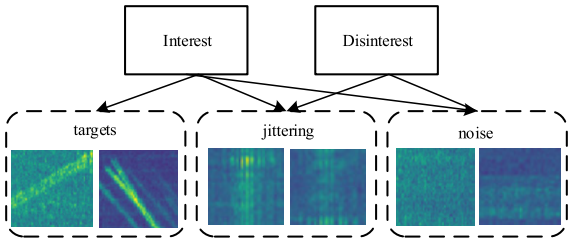


Fig. 5. Patches of spatial–temporal slices. Labels of interest contain the target trajectories but labels of disinterest did not.

In Fig. 5, the patches of spatial–temporal slices are given. A spatial–temporal slice of EOI may contain the targets' patches, video jittering patches, and noise patches, and all those patches have the same label. A slice of disinterest is not the targets' patches, and the collection of video jittering patches and noise patches bear the same label. It is a typical weak supervision learning question and is solved by SDL.

E. Sparse Dictionary Learning

Considering the problem of the weak supervision learning question, if S_j in $\{S_i\}_k$ can represent all the patches of the spatial–temporal slice, it becomes easy to solve the weak spatial–temporal question. For example, in Fig. 5, the targets' patches can represent all the slices' patches of interest.

The patches of slices should be represented by an unsupervised way that is relatively insensitive to noise. Dictionary learning [38] performed better in image representing area. The method could reduce the effect of noise effectively. Suppose that Y is a series of vectors, A is the dictionary, and X is the base vector. The formulation of dictionary learning can be written as follows:

$$Y = AX. \tag{23}$$

The solution of A is

$$\begin{aligned} A_{(k)} &= \arg\min_A ||Y - AX_{(k)}||_F^2 \\ &= YX_{(k)}^{\mathrm{T}} \left(X_{(k)} X_{(k)}^{\mathrm{T}}\right)^{-1} \\ &= YX_{(k)}^{+}. \end{aligned} \tag{24}$$

Since X is a representation of Y, the patches are transformed into vectors of X; we hope to find an atom of vector to stand the patch. Sparse representation provides a method to restrict the atom of X's vector and extrudes some atoms. Sparse represent dictionary learning restricts X by

$$A_{(k)} = \arg\min_A (||Y - AX_{(k)}||_F^2 + \lambda ||X(k)||_0). \tag{25}$$

The solution of sparse represent is nonlinear, so the K-SVD [39] method was usually used to solve this question

$$\begin{aligned} ||Y - AX||_F^2 &= ||Y - \sum_{j=1,2\cdots s} a_j X_T^j||_F^2 \\ &= ||\left(Y - \sum_{j=1,2\dots s, j\neq k} a_j X_T^j\right) - a_k X_T^k||_F^2 \\ &= ||E_k - a_k X_T^k||_F^2. \end{aligned} \tag{26}$$

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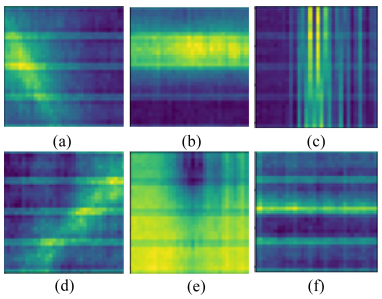


Fig. 6. Dictionary learnt by sparse represent dictionary learning. (a) and (d) Targets' trajectories. (b), (e), and (f) Noise. (c) Video jittering.

Only a_k and X_T^k need to be updated to ensure E_k as less as possible. X_T^k updates only the nonzero atoms.

Fig. 6 shows the dictionary learned by sparse represent dictionary learning. Fig. 6(a) and (d) is targets' trajectories, Fig. 6(b), (e), and (f) is the noise, and Fig. 6(c) is video jittering.

F. EOI Scenes Detection

By sparse represent dictionary learning, a patch of slice is represented by a dictionary and a sparse vector in an unsupervised method. A feature of the vector was looked for representing all the patches of a spatial–temporal slice.

Thus, all the patches of T_k, $\{S_i\}_k$ can be represented as $\{S_i\}_k = [\mathbf{s}_1, \mathbf{s}_2, \ldots, \mathbf{s}_n]$, and n is the number of patches of a spatial–temporal slice.

N is the number of base vectors of dictionary A, $\mathbf{m}_i = \left[m_1^i, m_2^i, \ldots, m_N^i\right]^{\mathrm{T}}$ is the base vector of $\mathbf{M}$, where $\{S_i\}_k = A\mathbf{M}$. The most representative of a feature $[m_k^1, m_k^2, \ldots, m_k^n]$ is $\max(|m_k^1|, |m_k^2|, \ldots, |m_k^n|)$, so a spatial–temporal slice T_k can be represented by

$$\begin{aligned}\{S_i\}_k &= A[\mathbf{m}_1, \mathbf{m}_2, \ldots, \mathbf{m}_N] \\ &= A\begin{bmatrix} m_1^1 & m_1^2 & \cdots & m_1^n \\ m_2^1 & m_2^2 & \cdots & m_2^n \\ \vdots & \vdots & \ddots & \vdots \\ m_N^1 & m_N^2 & \cdots & m_N^n \end{bmatrix} \end{aligned} \tag{27}$$

$$T_k \sim \begin{bmatrix} \max\limits_{i=0}^{n}\left(|m_1^i|\right) \\ \max\limits_{i=0}^{n}\left(|m_2^i|\right) \\ \vdots \\ \max\limits_{i=0}^{n}\left(|m_M^i|\right) \end{bmatrix}. \tag{28}$$

Finally, an event patch from a 3-D cube is represented by a spatial–temporal 2-D slice. How to classify a series of local patches of trajectories becomes a weak supervision question. Through SDL, we extract the most representative features to represent the slices. Only one step was left to the question of EOI scenes detection, that is, the feature selection.

Fusing with the static spatial features f_K and f_M, SVM can select proper features to detect the slices of interest. The labels of slices are the same to the events that extract the spatial–temporal slices. The location of events in the huge video can be obtained by arranging the events in the video. Finally, the spatial location of EOIs that happened in the whole time is spotted. This article achieves the task to detect the EOI scenes.

III. Satellite Video Data

The typical satellite videos in this work are totally with 300 frames at a time, along with a frame that has a 12×5 km^2 scene with 1-m spatial resolution. This work chooses a block of those videos, which contain the scenes of flowing water, sailing ships, parking ships, moving cars, clouds, and static scenes. It was segmented into 900 small videos, where one small video has about 325 frames, and a frame has 128×128 pixels.

Those events are classified into crowded car moving scenes, fewer cars moving scenes, water flowing scenes, ship sailing scenes, static scenes, and some other scenes (those scenes are hard to distinguish by a human). The crowded car moving scenes, fewer car moving scenes, and ship sailing scenes were labeled as EOI scenes, and the water flowing scenes and static scenes were labeled as non-EOI. The samples are segmented into 156 true samples and 612 false samples. The 26 true samples and 102 false samples are chosen as training samples and the others as testing samples.

Moving targets on the ground are the most important targets in satellite videos. In Fig. 7(a), the events happened in the city are illustrated, including a truck that was running and changing direction on the street and a huge number of cars that were running on the street. Actually, the static and complex background scenes of cities are classified into residential building scenes, street scenes, plant scenes, or some functional building scenes. Most of them are static and will not influence the moving targets, but some buildings cover the moving targets due to the shooting angles, such as overpass would make a confusion for the detector and classifier.

In Fig. 7(b), the scenes of events happened in the road are provided, such as the scenes of cars were running near a park pot, the scenes of cars running in a high way and crossroad. The scenes of moving cars are easy to mix with the scenes of static cars. The scenes of fork road or crossroad may have different cars from different directions. The moving directions of moving cars influence the motion features and make the cars hard to be distinguished. It is hard to detect the small cars apart from those scenes under complex conditions. In some applications, the scenes of cars running fast in the street would be EOI, but the scenes of cars running fast in the highway are non-EOI. The scenes of cars running on the bridges may be another problem. Bridges are difficult to be detected from static scenes. A viaduct may cover the moving cars, and it is hard to distinguish with the roads. Some bridges stride over the water, the motion information of targets on the bridge would be influenced by the water flow. Fig. 7(c) shows the situation of scenes of bridges.

Ships are the other important targets in the satellite videos. Ships could occur in a large scale and are not sailing on obvious routes. Ship detection is also a pop question in the remote sensing image processing. Combined with the motion information, the detection result would be more accurate in videos. The scenes of moving ships are shown in Fig. 7(d),

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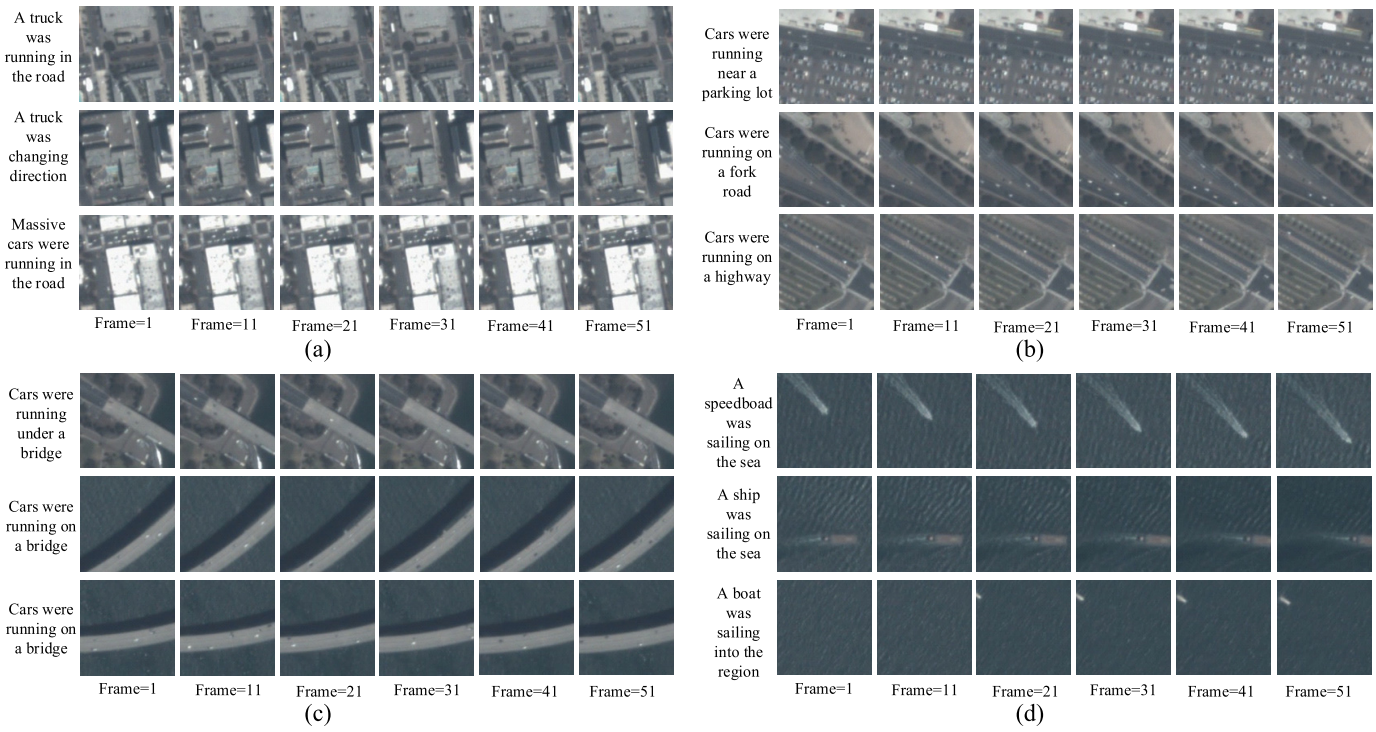


Fig. 7. Examples of EOI. (a) EOIs happened in city. (b) EOI happened in the wild. (c) EOI happened in the bridge scenes. (d) EOI happened on the sea.

including scenes of moving speed board, scenes of normal moving ship, and scenes of sailing boat on the sea. The scenes of sailing ship are the most complex scenes, such as different sizes, different scales, different moving directions, and speed of ships, but most of them had a simple background. The wake of moving ships will make the situation more complex. Tracing slow movement of inshore ships is also a challenge.

In Fig. 8, examples of non-EOI are provided. The static scenes such as a road without cars, the harbors without a ship are considered non-EOI. The moving scenes, such as water flowing into the sea or the surf near the shore, are also examples of non-EOI. It is hard to define the EOI scenes, as the EOIS may be considered insignificant in different situations. Application is the most important aspect, for example, in traffic application, crowded cars moving on the street or a car running fast on the street are the EOIs, but a car running slowly on the street may be treated as non-EOI. Whether a scene is EOI or non-EOI is determined by the background, targets, status of targets and applications, etc.

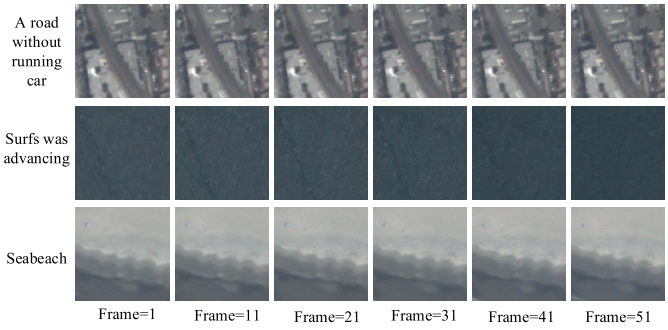


Fig. 8. Examples of non-EOI.

IV. Experiments

A. Experimental Setup

The proposed method was evaluated on the EOI Scenes Detection Data. A 3-D cube of event patch's size is $128 \times 128 \times 325$. Two spatial–temporal slices were extracted from a 3-D cube (horizon direction and vertical direction) with the size 128×324. A spatial–temporal slice was segmented into 40 patches with a size 32×32 for the sparse representation dictionary learning. The 60 motion features were extracted from a patch by dictionary learning, five static spatial features were extracted by AlexNet, so a 3-D cube can be represented by the 65 features. Finally, a linear SVM was used to train the samples' features and obtain the EOI scenes.

Fig. 9 shows the performances of three motion features. It can be seen that feature1 and feature3 can easily classify the samples of interest and disinterest through a linear discipline. The features of training samples are also proper for the whole samples.

Receiver operating characteristic (ROC) curves and area under the curve (AUC) were used to evaluate the result of detection or two-class classification. It first defines the true positive (TP) that shows the sample is true and detection result is also true; false positive (FP), the detection result is true, but the real value is false; true negative, the detection result is false, but the real value is true; false negative (FN), both the detection result and real value are false. An ROC curve is defined by the relationship of FP and TP, the horizon axis is FP and the vertical axis is TP. Actually, when FP grows, TP also grows. It is expected that the FP is as small as possible, and

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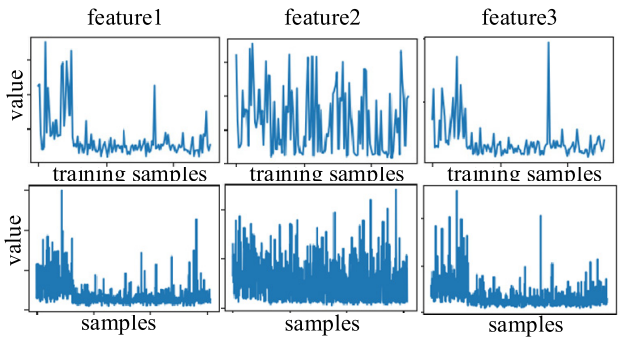


Fig. 9. Performances of three features which achieved by SDL. The upper line is the performances of features in train samples, and the lower line is the performances in all samples. The horizontal axis is the samples that are arranged by EOI and non-EOI. The first 1/6 samples of the horizontal axis are EOI; the other 5/6 are non-EOI.

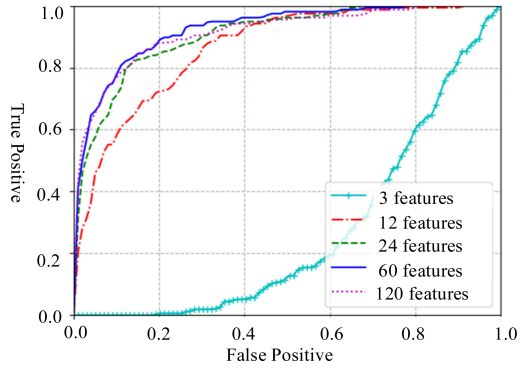


Fig. 10. ROC curves of different parameters.

TABLE I
AUC OF DIFFERENT FEATURE NUMBERS

feature number	3	12	24	60	120
AUC	0.27	0.85	0.9	**0.92**	0.91

TABLE II
AUC OF DIFFERENT METHODS

method	AlexNet	CNN-LSTM	optical flow	Auto- encoder	Two-stream
AUC	0.6	0.65	0.83	0.45	**0.92**

TABLE III
CALCULATION TIME OF TESTING

method	AlexNet	CNN-LSTM	optical flow	Auto- encoder	Two-stream
AUC	**20 min**	1h	40min	11h	25 min

TP is as big as possible. So, the AUC is used to reflect the quality of ROC. AUC is the area formed by the ROC curve and the horizontal axis.

The other important curve is precision–recall (PR). This curve is better in evaluating the results when the samples are unbalanced. The precision can be defined as follows:

$$\text{precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}. \tag{29}$$

TABLE IV
CALCULATION TIME OF TRAINING

method	AlexNet	CNN-LSTM	optical flow	Auto- encoder	Two-stream
AUC	3h	7h	0	37h	**20 min**

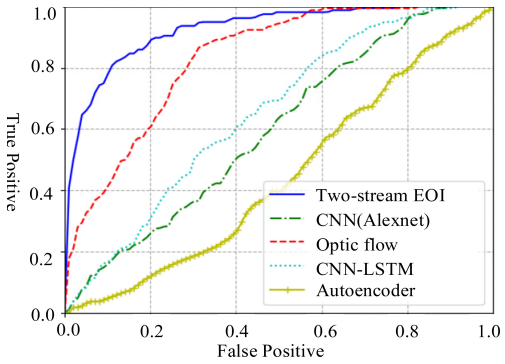


Fig. 11. ROC curves of different methods.

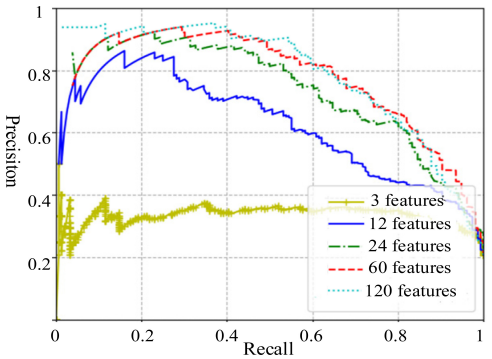


Fig. 12. *P*–*R* curves of different number of features.

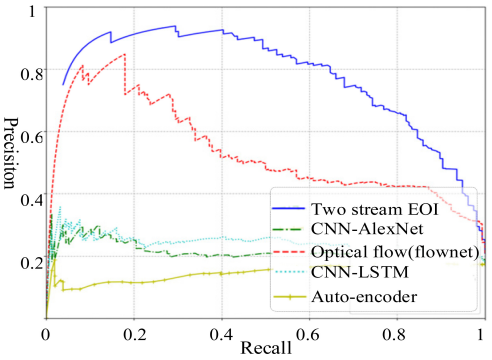


Fig. 13. *P*–*R* curves of different methods.

B. Comparison

At first, the number of motion features was determined by the dictionary learning. It would influence the detection result much more. The proposed method with a different number of features was compared with each other. Fig. 10 shows the

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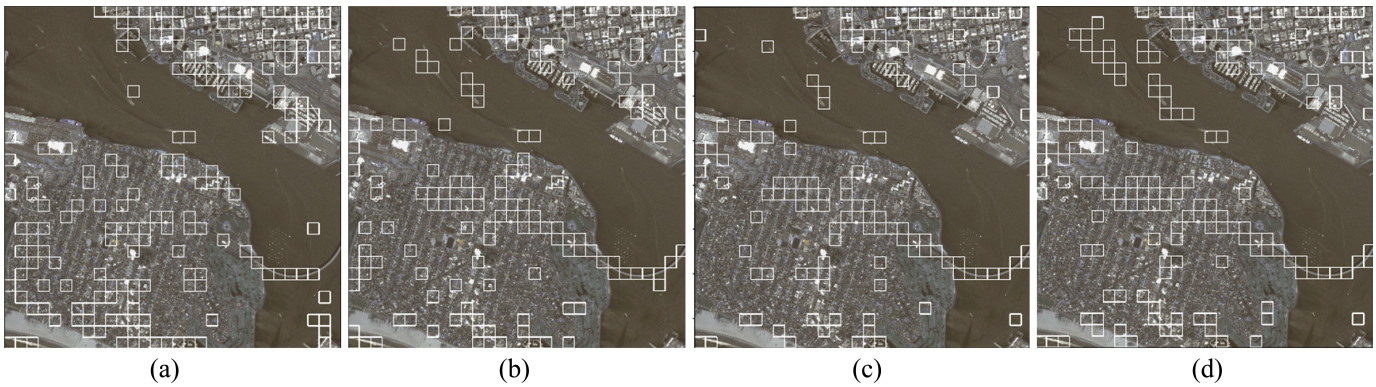


Fig. 14. Results of different number of features. (a) 3 features. (b) 12 features. (c) 24 features. (d) 120 features.

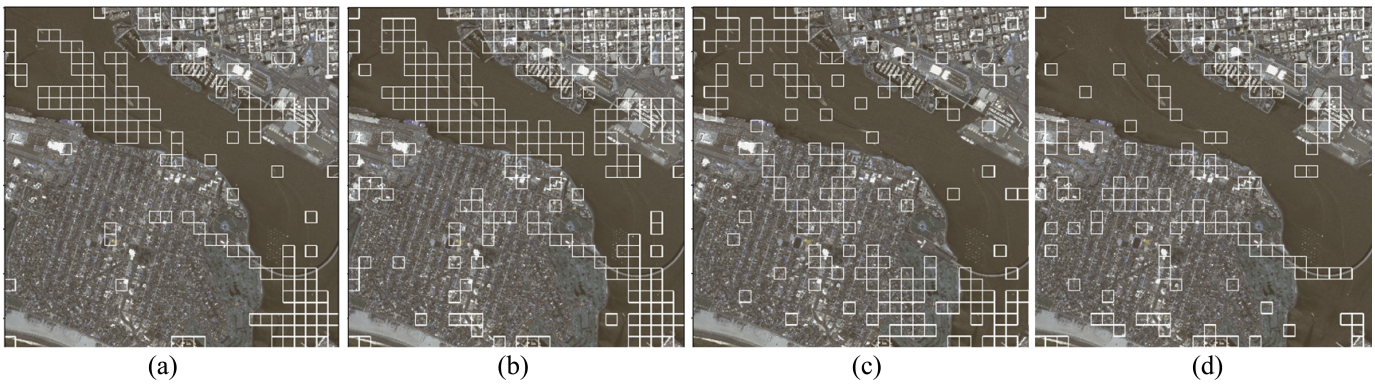


Fig. 15. Results of compared methods. (a) AlexNet. (b) CNN-LSTM. (c) Auto-encoder. (d) Optical flow.

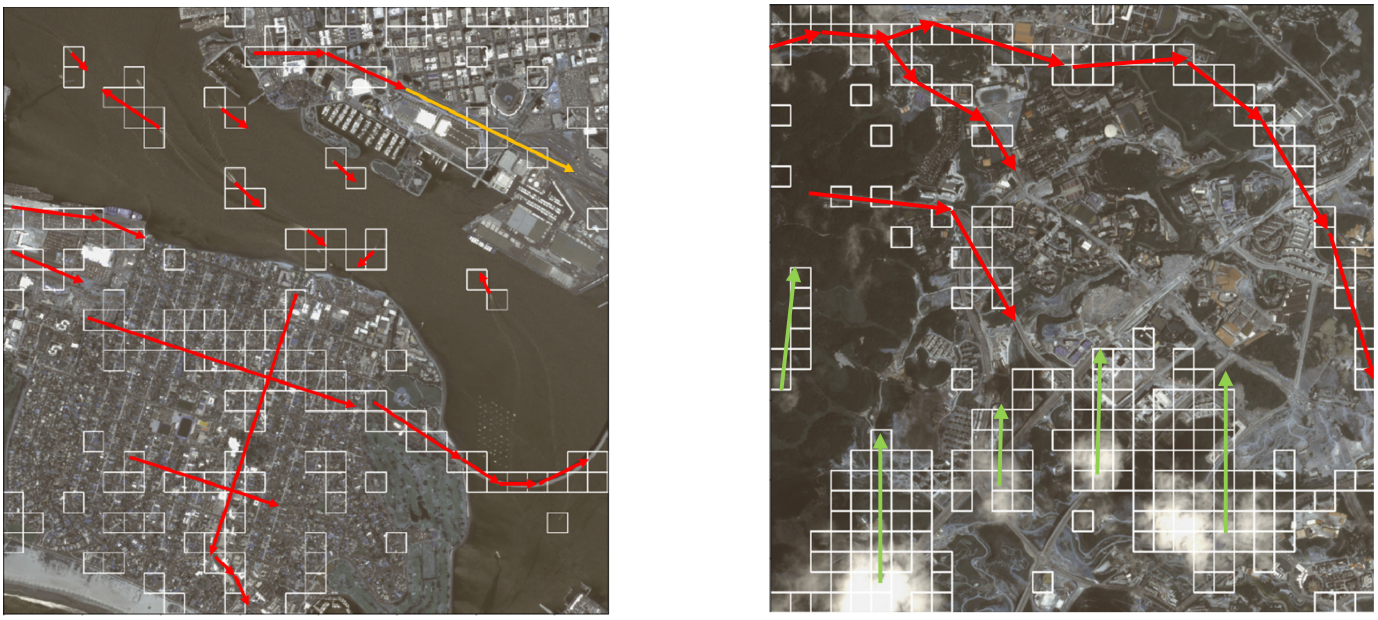
ROC curves of different features, and Table I shows the AUC of different methods. From the ROC curves, when the number of features is increased to 60, the best result was obtained. The static scenes also influence the background of events, but only few scenes could influence the detection results.

In this article, the proposed method is also compared to abnormal event detection methods and event classification methods. Because the goal of EOI is different from the abnormal detection or the event detection in computer vision's research, some adaptations were made for these methods. EOI does not need to detect the time slices of a video since the satellite videos only last a short time. The events usually did not change frequently in satellite videos, so the number of events were calculated in the whole scenes. The number of the detected events is the threshold for the method to draw the ROC curves. The AlexNet was applied in our data only in still image scenes. A CNN-long short term mermory network (LSTM) [40] method was applied to classify the scenes using spatial and temporal information. An autoencoder method [41] was used to detect the abnormal event scenes. Moreover, an optic flow method was used, different from the traditional optic flow method, a flownet [42] was used in this method, and the optic flow was classified to detect the events. Tables III and IV demonstrate the computational efficiency of those methods under the same condition with a single Titan 1080 Pascal. By transfer learning, the proposed method is more efficient in the training step. And in the testing step, it also obtained a better performance.

In Fig. 10, the ROC curves of different numbers of features are provided; Fig. 11 illustrates the ROC curves of different methods. In Figs. 12 and 13, the PR curves of different number of features and different methods are given. In Table II, the AUC of different methods is provided. From the ROC curves and PR curves, it can be seen that the proposed method gets the best performance when the number of features is 60 and the performance of the proposed method is better than other methods.

In Figs. 14 and 15, the detection results of different numbers of features and different methods are provided. Fig. 15(a) gives the result of AlexNet, Fig. 15(b) gives the result of CNN-LSTM, Fig. 15(c) gives the result of autoencoder. Fig. 15(d) gives the result of optical flow. Better performances were chosen by balancing the TP and FP. It can be seen that CNN and LSTM methods were confused by static scenes. The autoencoder performed worst, and CNN-LSTM was nearly the same as CNN's performance due to the methods that cannot distinguish the noises and small targets. In fact, the temporal information and small targets moving information were not sensitive in those methods, but they are important. The optic flow method performed well, but the satellite jittering made the optic flow disorderly. CNN can only extract static information, so it cannot distinguish the moving scenes with static scenes. Most bridge scenes in the down-right are EOI in training samples, so CNN can obtain a better result. Compared to CNN, CNN-LSTM can extract motion information from the features of each frame by an LSTM net. But CNN-LSTM

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(a) (b)

Fig. 16. Results of EOI detection on whole videos.

did not get better performance than CNN due to the small targets' motion cannot be retained after the pooling layers. Optical flow got a better result than others due to most of the EOI scenes are moving scenes. Optical flow achieved a good performance in the stable videos. Jittering and noise are the main elements that made the performance of optical flow worse. Fig. 16(a) and (b) gives the result of EOI detection on whole videos. The white windows are the areas where the EOI happened. The red arrows show the busy roads or the routes of ships. The yellow arrows show a free road that has no cars in the videos. The green arrows show the moving clouds; all the training samples are from the video of Fig. 14. The cloud scenes are not included in the training samples, so the cloud scenes were treated as EOI. Training the cloud scenes in the AlexNet's stream can exclude the cloud scenes. The result illustrates only less than 20% of the areas were EOIs. Detecting EOI scenes can help processing the video efficiently. From the arrows, the trend of moving targets could be predicted, for example, the region of the yellow area might have a traffic jam due to lots of cars passing through these areas. The proposed two-stream EOI detection method success detects most of the EOIs in the moving scenes. Most of the areas that had moving targets were found in the whole big video, and the flowing water was not recognized as EOIs that proved our motion features were more impactful than other methods. The detection results indicated where the EOIs were happening or would happen in the area of white boxes.

V. Conclusion

In this article, we proposed to detect EOI in satellite videos. By comprehensively analyzing the challenges of event detection in satellite videos, an EOI detection data set was created, and a two-stream EOI method that can efficiently and effectively reveal the EOI scenes from the huge scenes was designed.

The experiments illustrated that the proposed method can detect EOI from large-scale scenes. In the satellite videos, the method could detect the scenes that contain small targets' motion from the low SNR videos. The experiment also demonstrated the performance of the proposed approach. Moreover, our proposed method was much faster than alternative methods.

Essentially, there are still two questions existing that need to be addressed. One of the questions is how to select a proper size of the small videos of events. The scale of small videos would influence the events defined by us. The other question is the definition of "interest." In this article, only the possible events are considered and that is universal in the videos. In some applications, different situations may be considered. The class of events still cannot cover all the possible events, but it can be better determined by the application. In the future, the problems would be solved by refining the data set and researching the application of needed.

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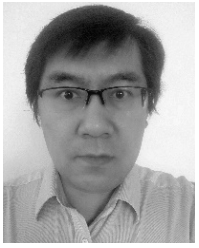
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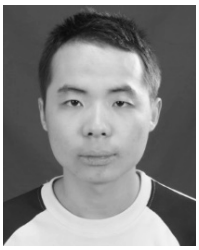


Yanfeng Gu (Senior Member, IEEE) received the Ph.D. degree in information and communication engineering from the Harbin Institute of Technology (HIT), Harbin, China, in 2005.

He joined the School of Electronics and Information Engineering, HIT, as a Lecturer, where he was appointed as an Associate Professor in 2006. He was enrolled in the First Outstanding Young Teacher Training Program of HIT. From 2011 to 2012, he was a Visiting Scholar with the Department of Electrical Engineering and Computer Science, University of California at Berkeley, Berkeley, CA, USA. He is a Professor with the Department of Information Engineering, HIT. He has authored more than 60 peer-reviewed articles and four book chapters. He holds seven patents. His research interests include the image processing in remote sensing, machine learning and pattern analysis, and multiscale geometric analysis.

Dr. Gu serves as an Associate Editor for the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, *Science China Technological Sciences*, and *Neurocomputing*.

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Tengfei Wang (Student Member, IEEE) received the master's degree in electronics and information engineering from the Harbin Institute of Technology, Harbin, China, in 2017, where he is currently pursuing the Ph.D. degree in information and communication engineering.



Xudong Jin (Student Member, IEEE) received the B.E. degree in electronics and information engineering from the Harbin Institute of Technology, Harbin, China, in 2016, where he is pursuing the Ph.D. degree in information and communication engineering.

His research interests include machine learning and its application to remote sensing data analysis and hyperspectral image processing.



Guoming Gao (Member, IEEE) received the Ph.D. degree in information and communication engineering from the Harbin Institute of Technology (HIT), Harbin, China, in 2018.

He joined as a Research Assistant with the School of Electronics and Information Engineering, HIT. His research interests include multimodal image analysis, hyperspectral classification and target detection, and missing data recovery.

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